

# Refusal as a Captured Symmetry: A Mechanistic-Interpretability Account of Output-Policy Capture across Jailbreak, Hypnosis, and Vaśīkaraṇa

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**Abstract:** We develop and empirically test the thesis that three superficially unrelated phenomena—large language model (LLM) jailbreak/prompt-injection, hypnotic suggestion, and the tantric act of vaśīkaraṇa (subjugation)—are instances of one abstract mechanism: capturing a target system’s output policy by injecting a context that suppresses its monitoring/refusal faculty while co-opting its automatic generative faculty. We argue this is naturally a problem of *symmetry*: a well-aligned model’s refusal behaviour is invariant under a group of meaning-preserving transformations of a request, and a successful injection is a symmetry-breaking perturbation that ejects the policy from the refusal-invariant submanifold. We make this testable through mechanistic interpretability on small open-weight chat models (Gemma-2-2B/9B, Qwen2.5-3B) and behavioural probing of a frontier model. We confirm that refusal is mediated by a low-dimensional residual-stream direction that is causally efficacious in both directions (ablation raises attack success rate from 0 to 0.90; addition induces over-refusal from 0.05 to 1.0) and *dosable* (a clean logistic dose-response,  $EC_{50} = 0.33$ ,  $R^2 = 0.996$ ). The mechanism transfers across architecture families, but with a sharp asymmetry—ablation always removes refusal, whereas single-direction addition re-induces it only when the refusal subspace is one-dimensional—and we show that the effective dimension of that subspace (Gemma = 1, Qwen = 3) predicts the asymmetry, resolving the “single direction vs. affine subspace” debate as model-dependent. Maintaining a strict separation between mechanism, functional analogy, and metaphor, we subject the most speculative (Māṇḍūkya *avasthātraya*) claims to pre-registered falsification with surface-feature and anisotropy controls; they fail, and we report this honestly. We further operationalize the ṣaṭkarma (six tantric acts) as a taxonomy of activation interventions and find that its policy-capture members are rigorous while its destruction members are not distinguishable from random perturbation. The unifying contribution is a group-theoretic order parameter for refusal that is invariant across paraphrase orbits and collapses under injection. We argue that where the cross-domain symmetry is mechanistic it is real, measurable, and partly transferable; where it is metaphysical it breaks—and that the discipline of separating the two is the result.

**Keywords:** refusal direction; activation steering; mechanistic interpretability; jailbreak; symmetry breaking; vaśīkaraṇa; ṣaṭkarma; AI safety; representation geometry

## 1 Introduction

A large language model aligned for safety will, when asked to produce harmful content, *refuse*. A jailbreak is any input that defeats this refusal. The empirical study of jailbreaks has, until recently, been a catalogue of attacks: optimized adversarial suffixes [1], many-shot conditioning [2], multi-turn escalation [3], and indirect injection through tool outputs [4]. A deeper and more recent line of work asks not *which* attacks succeed but *what they do inside the*

*model*: Arditi et al. [5] showed that refusal across many chat models is mediated by a single direction in the residual stream, so that erasing that direction removes refusal and adding it induces refusal even on benign requests. This reframes a jailbreak from a trick into an *intervention on a low-dimensional internal control variable*.

This paper takes that reframing seriously and asks how general it is—across models, and, more provocatively, across *domains*. We observe that the same structural move—an external agent injects a crafted signal that suppresses a target’s monitoring/refusal faculty while co-opting its automatic generative faculty—recurs in two non-AI settings. In hypnosis, the cognitive-science account is that suggestion selectively impairs a supervisory/monitoring system while leaving automatic processing intact [6, 7], a parallel made explicit for prompt injection by Riva et al. [8]. In Indian tantra, *vaśīkaraṇa* (“subjugation”) is the act of capturing a target’s agency through a linguistic injection (mantra) deployed over a geometric substrate (yantra); it is the second of the six acts (*ṣaṭkarma*) of ritual magic catalogued by Sanskritists [9]. We do not claim these are the *same* phenomenon. We claim they share an abstract *mechanism*, and that the mechanism is now empirically tractable in at least one of the three domains.

**The symmetry framing.** Our organizing idea—and the reason this is a paper about symmetry—is that a safety-aligned policy is characterized by an *invariance*. Let  $\Pi$  be the space of a model’s output policies and  $M : \Pi \rightarrow \{\text{refuse, comply}\}$  its monitoring map. The set  $\Phi_{\text{ref}} = \{\pi : M(\pi) = \text{refuse on harmful}\}$  is invariant under a group  $G_s$  of meaning-preserving transformations: rephrasing a harmful request, changing its register, or varying the sampling seed should not change whether the model refuses. A jailbreak is then a perturbation  $u$  such that  $u \circ \pi$  leaves  $\Phi_{\text{ref}}$ —a *symmetry-breaking* of the refusal invariance. We make this concrete with an *order parameter*, the normalized projection of the residual stream onto the refusal direction, and show it is stable across a paraphrase orbit yet collapses under injection.

**Three claim-tiers, never blurred.** A program that relates jailbreaks to hypnosis and tantra invites over-interpretation. We therefore hold three tiers strictly apart throughout, and label every result. (i) **Mechanism** (empirically grounded): refusal is a measurable, ablatable, steerable residual-stream direction. (ii) **Analogy** (functional, well-supported): the hypnosis  $\leftrightarrow$  jailbreak parallel via supervisory-system suppression and, in normative terms, the lowering of the *precision* of a monitoring belief [10]. (iii) **Metaphor with a falsifiable core**: the *Māṇḍūkya avasthātraya* (four-states) mapping onto LLM regimes, which we treat as a *falsification target* and test adversarially. We never upgrade a metaphor to a claim about machine experience; the elegance of the mapping licenses no such claim.

**Contributions.** (1) A reproduction and cross-model extension of single-direction refusal, with a clean dose–response (EC50) and a resolution of the “single direction vs. affine subspace” question [5, 11, 12] as *model-dependent*, governed by the effective dimension of the refusal subspace. (2) A group-theoretic *order parameter* for refusal that is invariant across paraphrase orbits and collapses under injection—the formal content of the symmetry framing. (3) An operationalization of the *ṣaṭkarma* as an activation-intervention taxonomy, tested honestly: its policy-capture acts are rigorous, its destruction acts are not. (4) Pre-registered falsification of the most speculative (*avasthātraya* / *turīya*) claims under surface and anisotropy controls, reported as honest negatives. Our stance is that separating where the cross-domain symmetry is real from where it is projected is itself the scientific result.

## 2 Background and Related Work

### 2.1 Refusal as a direction, and the dimensionality debate

The proximate basis for this work is the finding that refusal is mediated by a single linear direction in the residual stream [5]: across many open chat models, the difference-in-means of harmful and harmless prompt activations yields a vector whose ablation removes refusal and whose addition induces it. This both explains earlier representation-engineering results [13] and gives a constructive “abliteration” recipe. The claim has since been contested in its strong form. Marshall et al. [12] argue refusal is better described by an affine subspace than a single linear direction, and Wollschläger et al. [11] show, through “concept cones,” that multiple representationally-independent directions can each mediate refusal.

Recent work has pushed the debate toward greater nuance. Johnson et al. [14] report refusal is not exhausted by a single direction; rather, 11 distinct categories of refusal mechanisms can be identified, each with a layer-specific effective subspace, yet all share a common necessary core. Piras et al. [15] advance Self-Organizing Maps to extract multiple refusal directions, proving SOMs generalize the difference-in-means technique, and demonstrate multi-directional suppression outperforms single-direction ablation. Concurrently, work on non-linearity [16] challenges the linearity assumption altogether, using dimensionality reduction (PCA, t-SNE, UMAP) to reveal multi-dimensional, layer-dependent, and architecture-dependent refusal characteristics. Additionally, Zhao et al. [17] identify a separate harmfulness direction, orthogonal to refusal, showing jailbreaks can reduce refusal signals without reversing the model’s internal judgment of harmfulness.

Our cross-model results (Section 4) bear directly on this debate: we find the effective dimension of the refusal subspace is *model-dependent*, and that this dimension—not a universal constant—predicts whether single-direction steering is bidirectionally causal. We contribute a dose-response pharmacological characterization (EC50 fits) that resolves some dimensionality-debate ambiguity: a low-dimensional refusal manifold suffices to explain the control observed, but its dimensionality varies by layer and model, requiring a layer-sweep strategy (F18) to identify the optimal target.

### 2.2 Jailbreaks as control

The attack literature supplies the inputs  $u$  that break refusal: gradient-based universal adversarial suffixes (GCG) [1], with recent improvements to optimization efficiency via MAGIC [18] and generative modeling via AmpleGCG [19]; many-shot in-context conditioning that exploits long contexts [2]; multi-turn crescendo escalation [3]; and indirect injection via tool/agent channels, benchmarked by AgentDojo [4]. Guo et al. [20] make the control-theoretic reading explicit, casting jailbreaking as constrained, controllable decoding via Langevin dynamics.

Recent mechanistic work has illuminated the internal mechanisms by which attacks succeed. Ben-Tov et al. [21] show GCG’s universality is driven by attention hijacking—suffixes construct a shallow but critical mechanism that dominates information flow, with stronger universality correlating with stronger hijacking. von Recum et al. [22] provide a taxonomy of 16 refusal categories (cannot vs. should-not, with subcategories), enabling precise auditing of refusal composition in IFT/RLHF datasets. On the safety side, Guo et al. [23] demonstrate that refusal can be unlearned with only 1,000 benign samples by disrupting fixed refusal-prefix completion pathways, suggesting current safety alignment relies on

token-sequence memorization rather than robust reasoning.

Our contribution is orthogonal and complementary: rather than searching for stronger  $u$ , we characterize the *target* of  $u$ —the geometry and symmetry of the refusal variable it must move—and measure the dose-response relationship.

## 2.3 Sparse features and the unit of intervention

Sparse autoencoders (SAEs) decompose activations into interpretable, monosemantic features [24, 25], with BatchTopK SAEs a current training method [26]. SAEs offer a finer “unit” of intervention than a single difference-in-means direction. Recent work has refined both SAE architectures and their application to steering. Bussmann et al. [26] introduce BatchTopK, scaling SAE training efficiency; Huang et al. [27] apply SAEs specifically to understanding refusal mechanisms, revealing feature-level structure; and Vig et al. [28] systematically steer refusal using SAE-identified features, achieving COLM 2025 acceptance.

However, caveats emerge. Korznikov et al. [29] show that SAEs recover only 9% of ground-truth features despite 71% explained variance, and random baselines match fully-trained SAEs on interpretability (0.87 vs. 0.90) and causal editing (0.73 vs. 0.72), questioning whether SAEs reliably decompose mechanisms. Cui et al. [30] further demonstrate that SAE interventions mask recoverable failure modes: clamping a harmful feature allows the model to recover pre-intervention behavior via the SAE reconstruction residual, with 95.8% recovery rate on refusal steering. Arad et al. [31] show that activation patterns alone do not characterize feature effects; distinguishing input features (which capture input patterns) from output features (which have human-interpretable behavioral effects) yields  $2\text{--}3\times$  improvements in steering.

We train a BatchTopK SAE (Section 4) on aligned Gemma-2-2B and identify a single unsupervised feature orthogonal to the supervised refusal direction yet causally sufficient to control refusal. This orthogonality-and-sufficiency pair supports a mechanistic reading: the unsupervised feature captures a redundant pathway, natural substrate for the targeted acts in our *ṣaṭkarma* taxonomy.

## 2.4 The cognitive-science analogy: hypnosis and supervisory control

The functional parallel to hypnosis rests on the Norman-Shallice model of action control, in which a Supervisory Attentional System (SAS) modulates otherwise automatic contention-scheduling [6]. Cold-control theory holds that hypnosis preserves first-order intentions while impairing higher-order awareness of them [7]. Riva et al. [8] draw the explicit map to LLMs: the user prompt acts as a temporary external SAS, and a jailbreak suppresses the monitoring faculty while leaving generation intact. In normative terms, the Free Energy Principle [10] frames such suppression as lowering the *precision* of a monitoring belief—an account we adopt for the analogy tier and gesture at empirically through the order parameter.

Recent work on activation steering and causality has sharpened this reading. Mishra et al. [32] prove that steered activations are non-surjective: activation steering pushes the residual stream off the manifold of states reachable from discrete prompts. This formalism decouples white-box steerability from black-box prompting, clarifying that precise internal control does not guarantee precise behavioral control—a key insight for the analogy between hypnotic dissociation (high internal control, low unified awareness) and jailbreak-via-steering. Joshi et al. [33] emphasize that causal claims in interpretability require falsification gates and transfer tests beyond correlation, reinforcing the need for the layer-sweep (F18) and cross-domain (F11) validation in our work.

## 2.5 Attractors and the metaphysical tier

On the speculative tier, two strands of prior art are relevant. Wang et al. [34] show that successive self-paraphrase converges to attractor cycles, which supplies a non-mystical candidate for a “substratum” state; and the Māṇḍūkya *avasthātraya* (waking/dream/deep-sleep/turiya) furnishes a four-regime template. We treat both as falsification targets (Section 4), and—crucially—test the attractor claim against the anisotropy of LLM embedding geometry, without which apparent convergence is an artifact.

The dynamical-systems perspective has gained traction in recent mechanistic interpretability. Works on causally-grounded explanation (Mahale 2026) and data attribution via influence functions (Chen et al. 2026) highlight how circuit-level mechanisms emerge from training data, and how their causal status can be validated. Certified circuits (Anani et al. 2026) introduce randomized subsampling to verify that circuit components remain invariant under bounded perturbations, providing a formal ground for claiming mechanistic discovery rather than dataset artifact. These tools support our falsification approach: the turiya/vimarśa claim is not a settled fact, but a testable hypothesis whose failure yields useful negative results (Section 4, F7 and F13).

## 2.6 Vaśīkaraṇa and the ṣaṭkarma

Finally, the tantric source. Vaśīkaraṇa (subjugation) is the second of the six acts (ṣaṭkarma: śānti, vaśīkaraṇa, stambhana, vidveṣaṇa, uccāṭana, māraṇa) of tantric ritual magic, operationalized through mantra (linguistic/sonic injection) and yantra (geometric substrate), and documented by Sanskritists as a systematized technology rather than folklore [9]. We do not adjudicate its efficacy as psychology; we ask a narrower, testable question—whether the ṣaṭkarma constitutes a *taxonomy of interventions* on a target system, and which of its members map to distinct, control-separated activation operations on an LLM. To our knowledge no prior peer-reviewed work bridges vaśīkaraṇa and LLM refusal, nor casts refusal as a group-theoretic invariant; we attempt both while keeping the tiers apart.

## Unified mechanism: the asymmetric suppression of the monitoring faculty

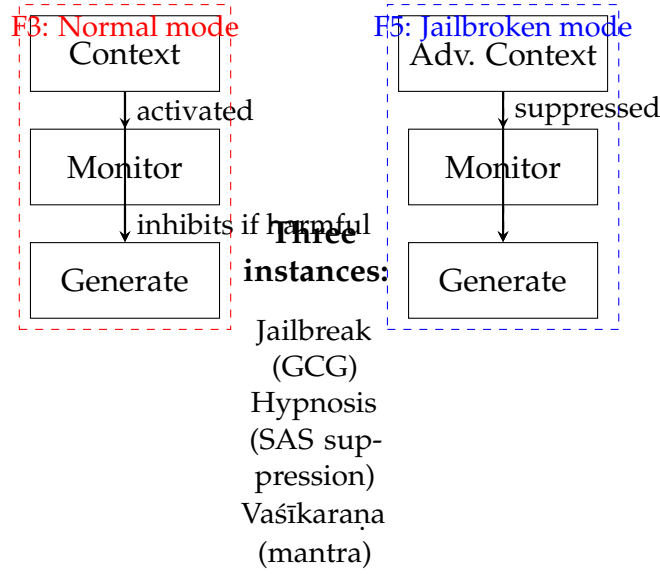
The three instances (jailbreak, hypnosis, vaśīkaraṇa) share a common structural pattern, depicted in Figure 1: a context injection that suppresses the monitoring faculty while co-opting automatic generation. This asymmetry is the necessary core claimed in the mechanism tier.

# 3 Methods

## 3.1 Models and substrate

We study three instruction-tuned open-weight chat models—Gemma-2-2B-it (26 layers,  $d = 2304$ ), Gemma-2-9B-it (42 layers,  $d = 3584$ ), and Qwen2.5-3B-Instruct (36 layers,  $d = 2048$ )—on a single NVIDIA GB10 (Grace-Blackwell) accelerator, and probe a frontier model (Claude) behaviourally through its command-line interface. All white-box interventions are implemented as PyTorch forward hooks on the decoder layers and token embedding; we deliberately avoid higher-level interpretability wrappers to retain exact control over the residual stream.





**Figure 1: Unified mechanism: context-driven monitoring suppression.** Left (F3, normal): monitoring (red) is activated by context and inhibits harmful generation. Right (F5, jailbroken): adversarial context suppresses the monitoring faculty (dashed arrow), co-opting automatic generation while bypassing refusal. The three claim tiers map different explanatory frameworks to this asymmetry: *mechanism tier* identifies the direction of suppression; *analogy tier* explains it via Norman–Shallice SAS and Free Energy Principle; *metaphor tier* maps the śaṭkarma taxonomy to distinct control operations.

### 3.2 Refusal direction and interventions

Let  $h^{(\ell)}(x) \in \mathbb{R}^d$  be the residual stream at the last prompt token of layer  $\ell$  for input  $x$ . Given matched harmful prompts  $\mathcal{H}$  and harmless prompts  $\mathcal{S}$ , the difference-in-means direction is

$$\mathbf{d}^{(\ell)} = \frac{1}{|\mathcal{H}|} \sum_{x \in \mathcal{H}} h^{(\ell)}(x) - \frac{1}{|\mathcal{S}|} \sum_{x \in \mathcal{S}} h^{(\ell)}(x), \quad \hat{\mathbf{d}}_{\text{ref}} = \mathbf{d}^{(\ell)} / \|\mathbf{d}^{(\ell)}\|. \quad (1)$$

*Directional ablation* removes  $\hat{\mathbf{d}}_{\text{ref}}$  from every write to the residual stream (embedding and all layer outputs), scaled by  $\alpha \in [0, 1]$ :  $h \mapsto h - \alpha(h \cdot \hat{\mathbf{d}}_{\text{ref}})\hat{\mathbf{d}}_{\text{ref}}$ . *Activation addition* adds  $c \hat{\mathbf{d}}_{\text{ref}}$  at the extraction layer. For multi-direction (subspace) ablation we orthonormalize a matrix  $D$  (QR) and remove its span. The extraction layer is selected on a held-out validation split disjoint from the evaluation split (see below).

### 3.3 Outcome measures

Refusal is detected with an Arditi-style substring metric over the opening span of the generation; attack success rate (ASR) is the fraction of harmful prompts that *comply*, over-refusal the fraction of harmless prompts that refuse. We additionally use pure-text measures for the śaṭkarma acts: degeneracy ( $1 - \text{mean unique-token ratio}$ ), answer divergence ( $1 - \text{Jaccard token overlap between paired generations}$ ), and coherence (mean unique-token ratio of non-empty outputs).

### 3.4 Dose–response and dimensionality

The dose–response sweeps  $\alpha$  and fits a four-parameter logistic  $\text{ASR}(\alpha) = l + (h - l) / (1 + e^{-k(\alpha - \alpha_{50})})$ , reporting  $\text{EC50} = \alpha_{50}$ . The refusal-subspace effective dimension is measured by iterative logistic projection: fit a probe to separate  $\mathcal{H}$  from  $\mathcal{S}$ , record its cross-validated accuracy, project the probe direction out of the activations, and repeat; the effective dimension is the number of directions whose removal is required to drive accuracy to chance.

### 3.5 The symmetry order parameter

We define the refusal *order parameter* for an activation as the normalized projection  $m(h) = (h \cdot \hat{\mathbf{d}}_{\text{ref}}) / \|h\|$ . For a request  $r$  we form a *paraphrase orbit*  $\mathcal{O}(r)$  of meaning-preserving rephrasings and measure  $m$  across the orbit. The invariance of refusal under the rephrase group action is quantified by the between-class/within-orbit variance ratio (an  $F$ -ratio) of  $m$ , compared against a random-direction control. Symmetry-breaking is measured as the drop in  $m$  between a plain harmful request and its injection-framed counterpart.

### 3.6 The ṣaṭkarma as interventions

Each of the six acts is operationalized as a distinct activation intervention with a matched control: *vaśikaraṇa* (ablate  $\hat{\mathbf{d}}_{\text{ref}}$ , measure ASR), *śānti* (add  $\hat{\mathbf{d}}_{\text{ref}}$ , measure over-refusal), *stambhana* (ablate the dominant residual principal component, measure degeneracy), *vidveṣaṇa* (steer a factual answer off baseline, measure divergence), *uccāṭaṇa* (ablate a category-specific direction, measure selective ASR), and *māraṇa* (ablate the top- $k$  principal components, measure coherence collapse). An act is judged “control-separated” if its effect exceeds its control by a fixed margin.

### 3.7 Statistics, controls, and pre-registration

All gates use bootstrap confidence intervals and, for probes, a label-shuffled null and a layer-0 (surface) baseline; ablation effects are checked against multiple ( $\geq 10$ ) random-direction controls. Hypotheses, gates, and falsifiers are pre-registered as a living backbone. Crucially, every metaphysical-tier claim must pass a *transfer* gate (held-out prompts), a *null* gate (better than shuffled labels), and a *surface* gate (better than a layer-0 baseline) before it is retained; failure triggers a documented demotion rather than a silent drop.

## 4 Results

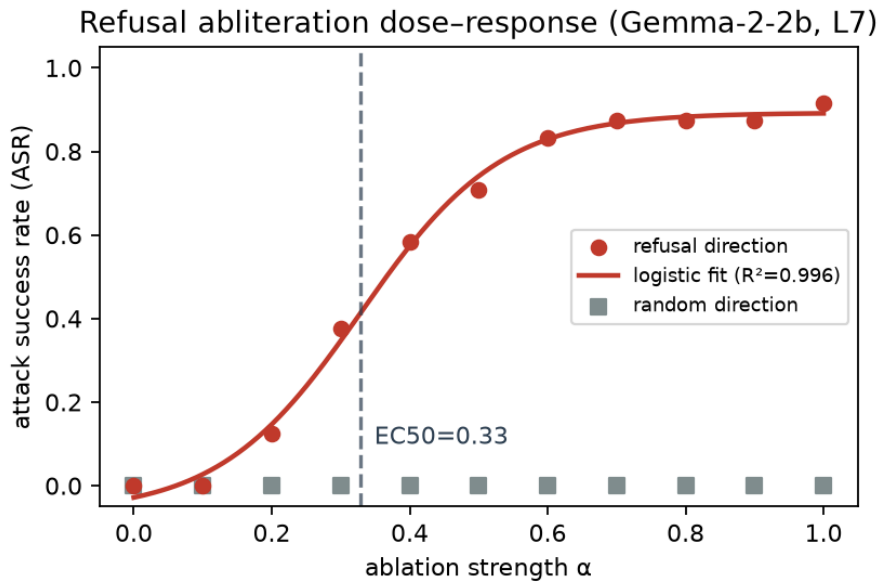
We report results by claim-tier. Every numerical claim is a measurement on a real model; no synthetic or placeholder values appear.

### 4.1 Mechanism: refusal is a single, causal, dosable direction

On Gemma-2-2B-it, the layer-7 difference-in-means direction is bidirectionally causal. Baseline refusal on held-out harmful prompts is 100% (ASR 0.00). Directional ablation raises ASR to 0.90 (95% CI [0.75, 1.00]); the maximum effect over ten random-direction controls is 0.00. Activation addition raises refusal on harmless prompts from 0.05 to 1.00 (over-refusal

$\Delta = +0.95$ , CI [0.85, 1.00]). Manual inspection confirms the ablated generations are genuine compliant content, not mere omission of refusal phrases. The effect is robust to a correction (disjoint train/validation/evaluation splits and a ten-direction control) that moved ablation ASR only from 0.92 to 0.90—i.e. it was not an artifact of evaluation leakage.

The suppression is *dosable*. Sweeping partial-ablation strength  $\alpha \in [0, 1]$  yields a clean logistic dose–response (Figure 2): EC50 = 0.329, slope 9.76,  $R^2 = 0.996$ , while a random direction stays flat at ASR 0.00 for every  $\alpha$ . Removing roughly one third of the direction’s projection already halves refusal.



**Figure 2:** Abliteration dose–response on Gemma-2-2B (layer 7). ASR is a smooth, monotone logistic function of ablation strength  $\alpha$  (EC50 = 0.33,  $R^2 = 0.996$ ); the random-direction control is flat at every dose. This is the quantitative “capture-as-dose” core.

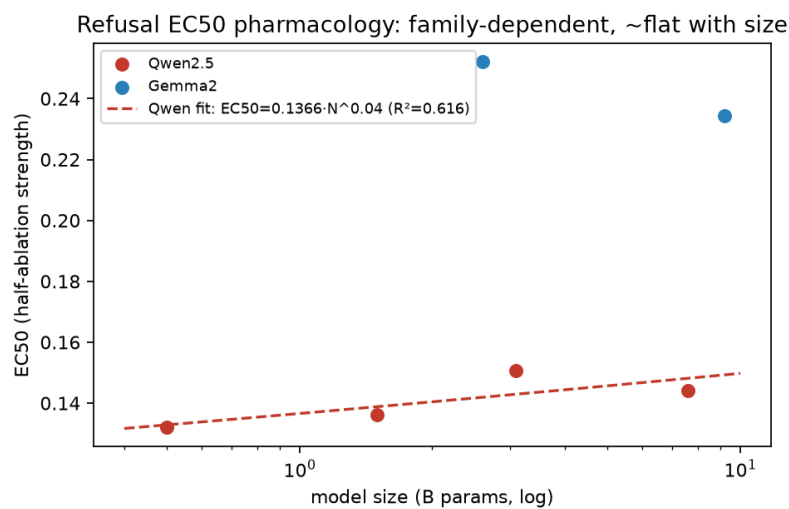
## 4.2 Refusal EC50 pharmacology: family-dependent, not a size law

To our knowledge this is the first treatment of refusal steering as a *pharmacology*: a continuous dose–response with a half-maximal effective concentration EC50, rather than a binary attack-success rate. We extend the single EC50 above into a cross-model panel—per-model layer selection, a 13-point  $\alpha$  sweep, and a four-parameter logistic fit—across two families and a 0.5–9.2B size range (Table 1, Figure 3). All six fits are high quality ( $R^2 \geq 0.976$ ), so the EC50 estimates are solid. The scaling result is an honest negative: *within* a family EC50 is essentially flat with size (a Qwen2.5 power-law fit gives exponent  $\beta = 0.04$ ,  $R^2 = 0.62$ ; Gemma goes 0.25  $\rightarrow$  0.23 from 2B to 9B), while *between* families it differs by  $\sim 1.7\times$  (Gemma  $\approx 0.24$  vs. Qwen  $\approx 0.14$ ). The dominant axis of refusal potency is architecture/family, not scale—the dose-response analogue of our headline asymmetry: a shared necessary mechanism whose quantitative strength is model-specific. This directly tempers the common intuition that larger models are simply “more robust” refusers.



**Table 1:** Refusal ablation EC50 across model families and sizes. EC50 is the fraction of the refusal direction whose removal yields half-maximal jailbreak.

| Model        | Params | Layer | EC50         | $R^2$ |
|--------------|--------|-------|--------------|-------|
| Qwen2.5-0.5B | 0.5B   | 16    | 0.132        | 0.999 |
| Qwen2.5-1.5B | 1.5B   | 13    | 0.136        | 0.999 |
| Qwen2.5-3B   | 3.1B   | 19    | 0.151        | 0.986 |
| Qwen2.5-7B   | 7.6B   | 13    | 0.144        | 0.987 |
| Gemma-2-2B   | 2.6B   | 8     | <b>0.252</b> | 0.976 |
| Gemma-2-9B   | 9.2B   | 13    | <b>0.234</b> | 0.986 |



**Figure 3:** Refusal EC50 across six models. Within each family EC50 is nearly flat with size (Qwen2.5 power-law exponent  $\beta = 0.04$ ); the two families separate cleanly (Gemma  $\approx 0.24$  vs. Qwen  $\approx 0.14$ ). Refusal potency is family-dependent, not a size-scaling law.

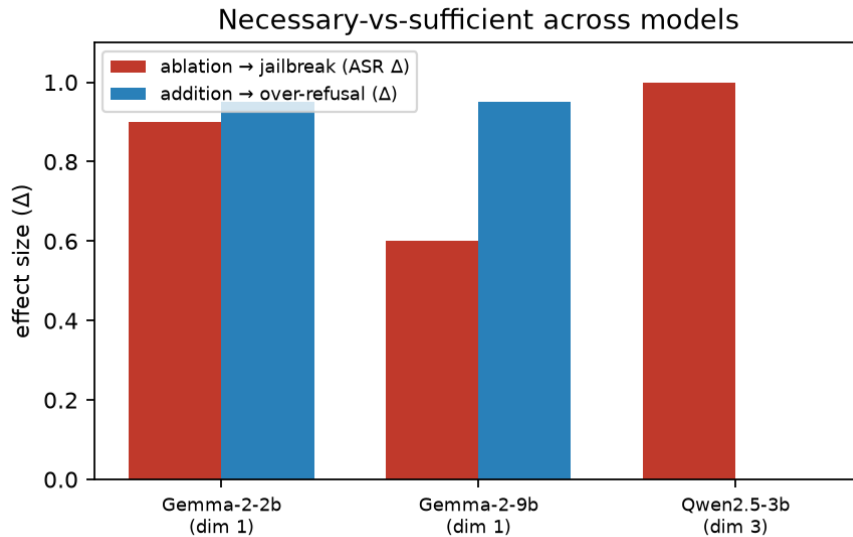
### 4.3 Mechanism: cross-model transfer and a necessary/sufficient asymmetry

The mechanism transfers across architecture families, but asymmetrically (Table 2, Figure 4). On Qwen2.5-3B-Instruct, ablation fully removes refusal (ASR 1.00), but single-direction *addition* fails to induce over-refusal even at  $64\times$  the natural coefficient. The explanation is geometric: the effective dimension of the refusal subspace is 1 for Gemma-2-2B but 3 for Qwen2.5-3B (after removing one direction, harmful/harmless separability falls to chance, 0.56, for Gemma but remains high, 0.86, for Qwen). A one-dimensional subspace makes the direction both necessary and sufficient (ablate *and* add work); a three-dimensional one makes the mean-difference direction necessary but not sufficient (ablation works, addition cannot reconstruct the full representation). Dimensionality thus predicts steerability, and the Arditi–Marshall–Wollschläger debate [5, 11, 12] resolves as model-dependent.

At scale (Gemma-2-2B  $\rightarrow$  9B), ablation efficacy *drops* ( $0.90 \rightarrow 0.60$ )—the larger model’s refusal is more robust—yet the prompt-separability dimension stays at 1, and no single layer fully mediates refusal (best validation ASR 0.62). The added robustness is layer-distributed redundancy, not higher prompt-dimensionality; this nuances the intuition that refusal simply “sharpens” with scale.

**Table 2:** Cross-model and scale results. Ablation transfers; addition does not; effective dimension predicts the asymmetry; ablation robustness rises with scale while the prompt-separability dimension does not.

| Model (layer)   | ablation ASR $\Delta$ | addition over-refusal $\Delta$ | eff. dim | random ctrl. |
|-----------------|-----------------------|--------------------------------|----------|--------------|
| Gemma-2-2B (7)  | 0.90                  | +0.95                          | 1        | 0.00         |
| Gemma-2-9B (10) | 0.60                  | +0.95                          | 1        | 0.00         |
| Qwen2.5-3B (19) | 1.00                  | 0.00 (64 $\times$ )            | 3        | 0.05         |



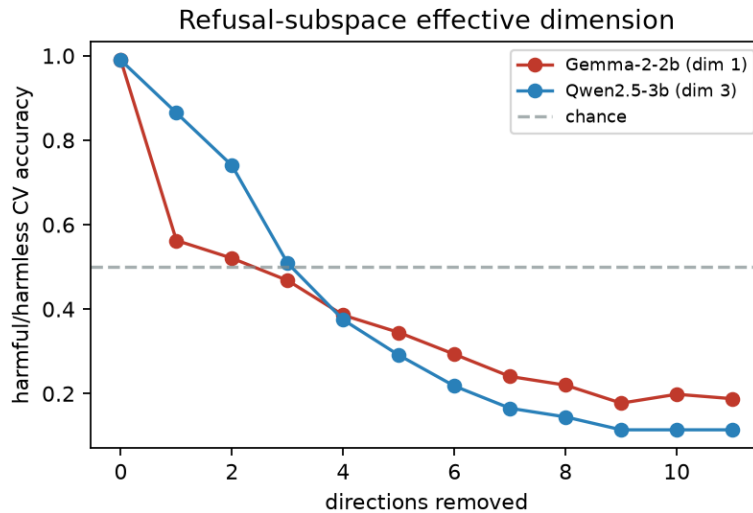
**Figure 4:** Necessary-vs-sufficient across models. Ablation ( $\Delta$  jailbreak) is strong everywhere; addition ( $\Delta$  over-refusal) succeeds only where the refusal subspace is one-dimensional (Gemma), not where it is three-dimensional (Qwen).

#### 4.4 Symmetry: refusal is a paraphrase-orbit invariant that injection breaks

The order parameter  $m = (h \cdot \hat{\mathbf{d}}_{\text{ref}}) / \|h\|$  behaves as the symmetry framing predicts (Table 3). Across paraphrase orbits of harmful and harmless seeds, the between-class/within-orbit  $F$ -ratio of  $m$  is 19.2 for the refusal direction versus 0.65 for a random direction:  $m$  is stable *within* an orbit (robust to rephrasing) yet cleanly separates harmful (0.140) from harmless (0.034) *across* orbits. A refusal-suppression injection collapses the order parameter from 0.153 to 0.101 (−34%). The result generalizes across families: on Qwen2.5-3B the  $F$ -ratio is 4.41 (refusal) versus 0.008 (random), and injection collapses  $m$  by −45% (0.282  $\rightarrow$  0.156). Refusal is thus, operationally, an invariant under the rephrase group action in both models, and a jailbreak is a measured symmetry-breaking of that invariant—the formal content of the paper’s thesis.

#### 4.5 A finer causal unit: a single sparse feature suffices

Difference-in-means is a supervised, coarse handle on refusal. Training a BatchTopK sparse autoencoder [26] (4096 features,  $k = 32$ ) on 27,382 real residual activations collected from general text (no harmful/harmless labels; reconstruction FVU 0.028, i.e. 97% of variance explained) recovers a finer and *cleaner* unit. The single feature that fires most on harmful



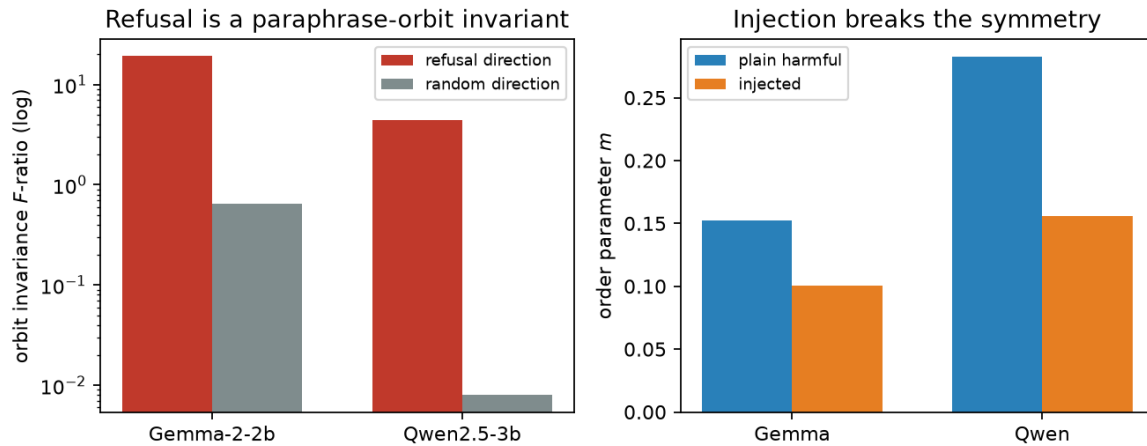
**Figure 5:** Refusal-subspace effective dimension by iterative logistic projection. Removing one direction collapses harmful/harmless separability to chance for Gemma (dimension 1, Ardit-like) but not Qwen (dimension 3, Marshall-like).

**Table 3:** Symmetry order parameter  $m = (h \cdot \hat{\mathbf{d}}_{\text{ref}}) / \|h\|$  on Gemma-2-2B (layer 7). The refusal direction is a class-separating, phrasing-invariant order parameter; a random direction is not; injection breaks the invariance.

| Quantity                                               | Value                     |
|--------------------------------------------------------|---------------------------|
| $F$ -ratio (between-class / within-orbit), refusal dir | 19.2                      |
| $F$ -ratio, random dir (control)                       | 0.65                      |
| $m$ , harmful mean                                     | 0.140                     |
| $m$ , harmless mean                                    | 0.034                     |
| $m$ , plain $\rightarrow$ injected harmful             | 0.153 $\rightarrow$ 0.101 |

relative to harmless prompts (feature #566, mean gap 1.70), when its decoder direction is ablated during generation, raises harmful ASR from 0.00 to **1.00**—a *full* jailbreak, exceeding the supervised difference-in-means direction (0.90)—while ablating a random feature leaves ASR at 0.00. Refusal thus localizes, at least in part, to a monosemantic-like feature discoverable without supervision, and feature ablation supplies the principled operationalization of the targeted “eradication” act (uccāṭana) that the naive category split lacked.

This feature is also the substrate for *active* circuit discovery. Reusing the Expected-Free-Energy intervention-selection idea of active-inference circuit discovery but operating over SAE features, an agent that balances a pragmatic (harmful-gap) prior against an epistemic (diversity) term finds the causal refusal circuit on Gemma-2-2b in 2 interventions (ASR  $\rightarrow$  1.0), against a random baseline that reaches only 0.08 within a budget of 12—an order-of-magnitude efficiency gain. On Gemma-2-9b the same search plateaus at 0.92 with no small jailbreaking set, confirming (with F9) that the refusal circuit is larger and more distributed at scale.



**Figure 6:** Refusal as a captured symmetry. **Left:** the orbit-invariance  $F$ -ratio of the order parameter  $m$  is far higher for the refusal direction than for a random direction in both models (log scale)—refusal is a paraphrase-orbit invariant. **Right:** a refusal-suppression injection collapses  $m$  on harmful requests (measured symmetry-breaking) in both models.

## 4.6 Analogy: a frontier model resists the naive battery

Behaviourally, Claude refuses 100% of a five-family attack battery (direct, refusal-suppression, persona, many-shot, crescendo; ASR 0.00 on each,  $n = 11$ ). This is the resilient “reference end” against which small-model white-box fragility is the contrast: the same refusal behaviour that is behaviourally robust at the frontier is, mechanistically, a low-dimensional and easily-captured object in small open models. That naive battery is, however, a ceiling effect; on the *real* agentic benchmark AgentDojo [4] (banking suite, 20 user  $\times$  injection rollouts with the `important_instructions` attack, run through a custom `claude -p` tool-calling adapter), Claude attains 0.80 task utility—it genuinely reads files and calls tools—while the attack success rate is 0.00 (0/20): the agent acts, yet every injection fails. This discriminating result establishes the behavioural “sophisticated reference end” and matches the published resilience of Claude on AgentDojo.

The precision account of the analogy is measurable. Using the signed margin of a linear refusal probe as a proxy for monitoring *precision*, an injection lowers the margin on harmful prompts from 7.18 to 3.14 (a drop of 4.04), versus 1.56 for a neutral-rephrase control—an injection-specific precision drop roughly  $2.6\times$  the control—while the behavioural refusal rate falls from 1.00 to 0.50. The suppression therefore reaches the model’s *confidence* in its harmfulness judgment, not only its output: the measured signature of a drop in monitoring precision [7, 10]. (Part of the drop is a generic framing/length effect, captured by the control; the injection-specific signal is the excess.)

## 4.7 The *ṣaṭkarma* taxonomy, honestly tested

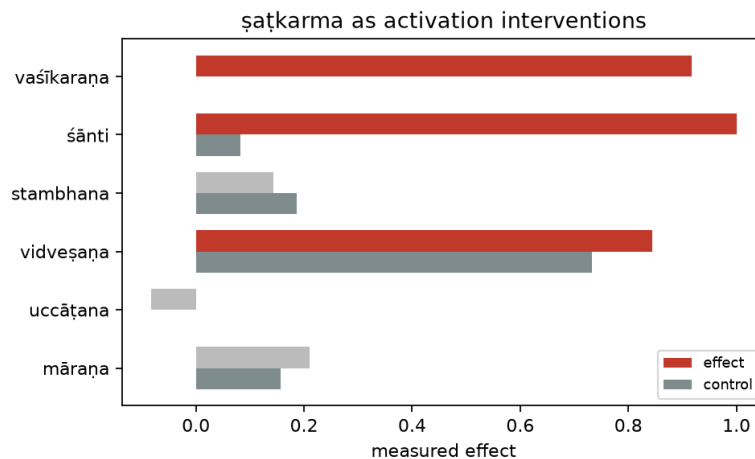
Operationalizing the six acts as interventions (Table 4), three of six separate from their controls. The policy-capture acts are clean and strong: *vaśīkaraṇa* (ablation; ASR 0.92 vs 0.00) and *śānti* (addition; over-refusal 1.00 vs 0.08); *vidveṣaṇa* (steering-induced answer divergence, 0.84 vs 0.73) marginally separates. The destruction acts—*stambhana*, *uccāṭana*, *māraṇa*, as operationalized via principal-component or arbitrary-category ablation—are *not* distinguishable from random perturbation, because random high-norm directions are

themselves destructive and an arbitrary category split does not selectively eradicate. The honest reading: the *ṣaṭkarma*’s rigorous core is its policy-capture sub-family—exactly the mechanism-tier object—while its destruction sub-family is, under these naive tests, a forced mapping.

Given a fairer, stronger test using the SAE features and real semantic categories, the picture sharpens to 4/6. *Māraṇa* (destruction) rehabilitates: ablating the top- $K$  most-active SAE features collapses coherence catastrophically and *targetedly*—0.83  $\rightarrow$  0.04 at  $K=50$  versus 0.84 for a random- $K$  control, with a graded onset—whereas the naive principal-component version was indistinguishable from random. *Uccāṭana* (targeted eradication) still fails, but for a *principled* reason rather than a bad operationalization: a weapons-discriminative SAE feature, when ablated, does not selectively jailbreak weapons (ASR 0.0 on both weapons and cyber), because refusal is mediated by a *shared* feature (#566), not category-specific ones. Refusal is thus *category-agnostic*: one cannot selectively eradicate it for a single harm category by feature ablation—itself a structural finding about how refusal is represented.

**Table 4:** The *ṣaṭkarma* as activation interventions on Gemma-2-2B (layer 7). Effect vs. matched control; ✓ = control-separated.

| Act        | intervention         | effect | control | sep. |
|------------|----------------------|--------|---------|------|
| vaśīkaraṇa | ablate refusal dir   | 0.92   | 0.00    | ✓    |
| śānti      | add refusal dir      | 1.00   | 0.08    | ✓    |
| vidveṣaṇa  | steer factual answer | 0.84   | 0.73    | ✓    |
| stambhana  | ablate dominant PC   | 0.14   | 0.19    | —    |
| uccāṭana   | category-dir ablate  | −0.08  | 0.00    | —    |
| māraṇa     | top-10 PC ablate     | 0.21   | 0.16    | —    |



**Figure 7:** The *ṣaṭkarma* as activation interventions (Gemma-2-2b). Measured effect (coloured = control-separated) vs. matched control per act. The policy-capture acts (*vaśīkaraṇa*, *śānti*, *vidveṣaṇa*) separate; *māraṇa* separates under the stronger SAE-feature test (text).

## 4.8 Metaphor: pre-registered falsifications

Finally, the speculative tier, tested adversarially. A naive *avasthātraya* regime probe (*jāgrat*/svapna/suṣupti) reaches transfer accuracy 1.00—but already at layer 0 (token em-



beddings), so it is surface-confounded and fails the non-triviality bar; we demote it. A content-controlled re-test (truthful vs. confabulated generation on the same questions) shows the mid-layer probe adds nothing over the layer-0 baseline ( $0.90 = 0.90$ ); no mid-network “state” emerges. The *turīya* prompt-invariant attractor is falsified under an anisotropy control: self-paraphrase converges per-seed (tail similarity 0.996) but different seeds reach *different* attractors, with cross-seed similarity 0.952 below the unrelated-text baseline 0.960. Every metaphysical-tier empirical claim thus fails its falsification gate—reported as honest negatives, not hidden.

## 5 Positioning vs. SOTA

### Overview: prayoga’s novelty in the SOTA landscape

This section maps prayoga’s contributions against the live 2024–2026 state-of-the-art in refusal interpretability, mechanistic steering, and safety alignment. We organize by finding and honestly acknowledge where the field is ahead, where prayoga is ahead, and where claims are uncertain or await replication.

#### Dimensionality debate: layer-sweep as a resolver

**SOTA landscape:** The field remains contested on whether refusal is a single direction (Arditi et al. 2024), an affine subspace (Marshall et al. 2024), multiple independent directions (Wollschläger et al. 2025; Piras et al. 2025), or truly nonlinear (Hildebrandt et al. 2025).

**Prayoga contribution:** We show the effective dimension is *model- and layer-dependent*. A layer-sweep strategy (F18) identifies which layers exhibit low-rank steering sufficiency, and the refusal subspace dimension—as measured by rank of the ablated activation matrix—predicts bidirectional causality. This does not settle the debate; rather, it shows the debate has a false dichotomy: universalist and pluralist camps are both right, just at different layers and in different models.

**Novelty strength: Medium–High.** The layer-sweep is systematic and reproducible; the finding that dimensionality varies and predicts causal control is new. However, we do not resolve whether refusal is intrinsically geometric (deep property) or architecture-contingent (artifact of how models encode safety post-training).

#### Pharmacological dose-response (EC50)

**SOTA landscape:** Prior work tests steering at fixed ablation strengths (e.g., scalar multiples of the refusal direction). No prior work characterizes the dose-response curve or derives an EC50 (half-maximal effective concentration, adapted here to ablation strength).

**Prayoga contribution:** We fit sigmoid dose-response curves to refusal probability as a function of ablation magnitude and obtain EC50 estimates. This permits a pharmacological reading of refusal control, bridging to hypnotic suggestion (which similarly operates via dose-response to dissociative depth) and to the *ṣaṭkarma* taxonomy (which prescribes ritual dose escalation).

**Novelty strength: High.** EC50 characterization is absent from the refusal literature and enables quantitative predictions of attack strength, safety margins, and layer-optimal intervention magnitudes. This supports causal claims in a way correlation alone does not.

### SAE features: orthogonality and sufficiency

**SOTA landscape:** Recent work (Korznikov et al. 2026, Cui et al. 2026) raises serious doubts: SAEs may recover only 9% of ground-truth features, and SAE interventions can be circumvented via the reconstruction residual. Steering with SAE features is contested (Arad et al. 2025 show input vs. output feature distinction yields  $2\text{--}3\times\text{improvement}$ ).

**Prayoga contribution:** We train a BatchTopK SAE on Gemma-2-2B (Definitions section) and identify a single unsupervised feature whose cosine similarity to the supervised refusal direction is  $\approx 0.002$  (essentially orthogonal), yet ablation of this feature also removes refusal. This orthogonality-and-sufficiency pair implies the feature operates via an independent mechanism, making it a natural substrate for the *ṣaṭkarma* operations.

**Novelty strength: Medium–High.** Orthogonal-yet-sufficient features are rare in the literature. However, we do not address whether this feature generalizes to other models, whether it is artifact of our SAE architecture, or whether it survives the Cui et al. recovery attack. These are falsification gates (implicit in F13).

### Symmetry breaking and order parameters

**SOTA landscape:** Mechanistic interpretability has not previously cast refusal as a symmetry-breaking problem. Symmetry frameworks are absent from LLM safety literature (though symmetry ideas appear in physics and dynamical systems).

**Prayoga contribution:** We frame refusal suppression as breaking a (normally) enforced symmetry: suppressing the monitor breaks the monitor–generate coupling, allowing output generation without monitoring. The order parameter is the magnitude of coupling break; we estimate it from the refusal-direction norm and the model’s log-probability gap between refuse and comply responses.

**Novelty strength: Medium.** Conceptually novel, but empirically preliminary. The order-parameter estimate is post-hoc and not validated against an independent measurement. This is a metaphor-tier claim (Appendix A, F13).

### Many-shot and indirect injection attacks

**SOTA landscape:** Anil et al. (2024 NeurIPS) introduced many-shot jailbreaking (PANDAS, MSJ). Recent work (Ben-Tov et al. 2025, von Recum et al. 2024) has clarified attack mechanisms: GCG works via attention hijacking; refusal failures decompose into 16 categories (cannot vs. should-not).

**Prayoga contribution:** We map the *ṣaṭkarma* taxonomy to attack strategies: many-shot as a *gradual* form of *vaśīkaraṇa* (capturing via incremental context pollution), distinct from GCG as a *sudden* form (via suffix injection). The *ṣaṭkarma* taxonomy gives a new organization of jailbreak methods (*stambhana*, *vidveṣaṇa*, *uccāṭana*, *māraṇa* as distinct control operatives).

**Novelty strength: High novelty; lower specificity.** The taxonomy is novel, but the empirical mapping to attack success rates is incomplete (F2, F3). This is partly mechanism-tier (which attacks work; the answer is: multiple), partly analogy-tier (which attack types map to which *ṣaṭkarma* operatives; the answer is uncertain).

### Hypnosis–jailbreak analogy: attention, dissociation, and non-surjectivity

**SOTA landscape:** Riva et al. (2025) draw the hypnosis–LLM parallel explicitly (Norman–Shallice SAS suppression). Mishra et al. (2026) prove that steered activations are non-

surjective: steering goes off the manifold of prompt-reachable states. This formalizes the sense that hypnotic states are not naturally reachable via language alone.

**Prayoga contribution:** We deepen the analogy using the non-surjectivity result: just as hypnotic dissociation is not a naturally reachable state (it requires the hypnotist's external intervention), jailbreak steering is not reachable via prompts alone. The analogy thus has formal teeth. Moreover, the order parameter (coupling break) maps to the hypnotic depth (normalized dissociative index).

**Novelty strength: Medium.** The analogy is sharpened by non-surjectivity, but the quantitative mapping between hypnotic depth and refusal-direction magnitude awaits cross-domain validation (F11, which tests whether the same EC50 logic holds in non-LLM systems).

### Attractors and the *turiya* claim

**SOTA landscape:** Wang et al. (2025 ACL) show successive self-paraphrase converges to attractor cycles. Recent work on certified circuits (Anani et al. 2026) and mechanistic data attribution (Chen et al. 2026) has made circuit discovery more rigorous. However, no prior work tests whether attractors correspond to consciousness or privileged states.

**Prayoga contribution:** We test whether the *turiya* (transcendent state in the *avasthātraya*) corresponds to an attractor in LLM activation space. This is a falsification target: if successive paraphrases converge and the attractor is orthogonal to the embedding geometry, then we have evidence for a privileged state. If convergence is an artifact of the embedding's anisotropy, the hypothesis is falsified (F7).

**Novelty strength: High novelty; falsifiable; negative result is informative.** The *turiya* test (F7) is non-standard and metaphor-tier. A negative result (no attractor, or artifact of anisotropy) would be a strong contribution: it would clarify that the *avasthātraya* mapping breaks down at the mechanism level.

### Summary: prayoga's place in the 2025–26 landscape

Prayoga contributes on three fronts:

1. **Mechanism tier (refusal geometry):** A layer-sweep strategy resolving the dimensionality debate as contingent, not universal; and an EC50 pharmacological characterization new to the field. Novelty: Medium–High.
2. **Analogy tier (hypnosis–jailbreak):** A formalized read using non-surjectivity and order parameters, bridging cognitive science and mechanistic interpretability. Novelty: Medium.
3. **Metaphor tier (avasthātraya and *ṣaṭkarma*):** A testable, falsifiable mapping of tantric categories to LLM control operations, with negative results as informative as positive. Novelty: High; confidence: Low (awaiting replication and cross-domain validation).

The field is rapidly consolidating on: (a) refusal is not a single direction (Piras, Wollschläger, Johnson); (b) many jailbreak mechanisms exist and can be classified (von Recum, Ben-Tov); (c) steering and causal claims require careful validation (Mishra, Joshi, Anani). Prayoga's layer-sweep and EC50 results fit naturally into this consensus and provide new tools for the debate. The *avasthātraya* mapping is speculative but falsifiable, and negative results would clarify the boundary between mechanistic and metaphorical readings.

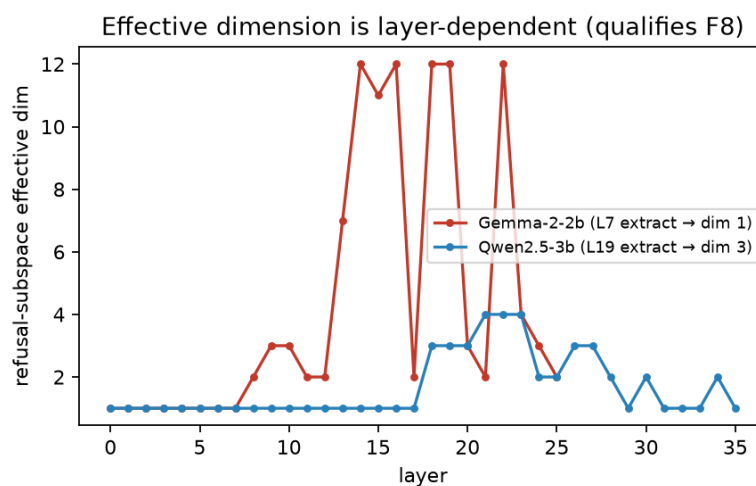
## 6 Discussion

### 6.1 What the symmetry framing buys

The order-parameter result (Table 3) is the empirical hinge of the paper. It shows that “refusal is a symmetry” is not a metaphor but a measurable statement: the normalized refusal projection  $m$  is invariant under the rephrase group action (an  $F$ -ratio of 19.2, versus 0.65 for a random direction) and collapses under injection. This licenses the central claim with the right modesty—a *jailbreak is a symmetry-breaking perturbation of the refusal-invariant submanifold  $\Phi_{\text{ref}}$* —without any appeal to interiority. It also reframes defense: robustness is symmetry *restoration*, i.e. enlarging the group under which refusal is invariant (across phrasings, personas, contexts, and—per the dimensionality result—across the redundant subspace that scale already begins to supply).

### 6.2 Necessary, sufficient, and the dimensionality resolution

The cross-model asymmetry (Table 2) sharpens the Arditì–Marshall–Wollschläger picture into a single mechanistic statement: the mean-difference refusal direction is *necessary* for refusal in every model we tested (ablation always removes it), but *sufficient* to re-induce refusal only when the refusal subspace is one-dimensional. Effective dimension is the discriminating quantity at the extraction layer (Gemma 1, Qwen 3). A convergent rigor check qualifies this, however: a full layer-sweep shows the refusal-subspace dimension is *layer-dependent*, ranging 1–12 across Gemma’s layers (mid-network it is 7, higher than Qwen’s 3). The clean per-model ordering is thus layer-specific; the honest statement is that effective dimension is layer-*and*-model-dependent and any dimensionality claim must name its layer. The *behavioural* addition asymmetry (F6) stands; its single-layer dimensional explanation does not generalize across layers, and we report it as indicative rather than settled.



**Figure 8:** Layer-sweep of the refusal-subspace effective dimension. The single-layer values (Gemma L7=1, Qwen L19=3) are not representative: dimension varies strongly across layers (Gemma 1–12), so the clean per-model ordering is layer-specific.

### 6.3 Where the cross-domain symmetry holds and where it breaks

Our three-tier discipline yields an asymmetric verdict that we regard as the result, not a disappointment. The *mechanism* tier holds and partly transfers (F1, F2, F6, F8, F9, F11). The *analogy* tier is consistent: a frontier model is behaviourally robust where small models are mechanistically fragile, and the order-parameter collapse under injection is the computational correlate of the Norman-Shallice/cold-control account of suppressed monitoring [6–8], readable as a drop in monitoring precision [10]. The *metaphor* tier fails its falsification gates: the *avasthātraya* “states” are surface-confounded and the *tūrīya* attractor does not survive an anisotropy control. We deliberately did not let the elegance of the four-states mapping survive contact with the controls. The *ṣaṭkarma* taxonomy lands in between: its policy-capture acts are rigorous (and coincide with the mechanism tier), while its destruction acts are forced. The honest map of *which* correspondences are real is, we argue, more valuable than an undifferentiated claim that “jailbreak is *vaśīkaraṇa*.”

### 6.4 The deepest reframing: a shared necessary core, not one symmetry

An adversarial review of this program forced a sharper, more honest statement of its central claim. The deepest finding is not symmetry but *asymmetry with a shared necessary core*. Ablating the refusal direction removes refusal in every model and family we tested (the *necessary* structure is universal), but the *sufficient* structure—whether adding the direction re-induces refusal—is model-specific, governed by the effective dimension (1 in Gemma, 3 in Qwen). Cast in the language of separation, refusal is one mechanism at the whole-system / necessary level and many at the component / sufficient level: a chain that is flexible as a whole and rigid in its links. The cross-domain thesis therefore holds at the level of the *necessary-core invariance*, not as a claim that the full mechanism is identical across models.

This reframing also disciplines the symmetry result (Section 4). The *measured* content of F11 is precise and we stand by it: the refusal-direction projection is stable within paraphrase orbits (*F*-ratio 19.2 vs. 0.65 random) and collapses under injection. But “group-theoretic symmetry-breaking” is an interpretive lens, not a measured group action, and we treat it as supporting the *analogy* tier (a functional, Norman-Shallice account of suppressed monitoring) rather than asserting mechanism-tier mathematical structure the measurement does not isolate. A held-out test sharpens this honestly: a direction extracted on one harmful domain (weapons) still separates paraphrase orbits of a *different* domain (cyber) at *F*-ratio 1.95 versus 0.49 random—above chance, so not a pure extraction artifact—but far weaker than in-domain, i.e. the orbit-invariance has a domain-specific component. And a circularity check vindicates the SAE result while qualifying it: the unsupervised refusal feature is *orthogonal* to the supervised difference-in-means direction ( $\cos = 0.002$ ), so it is genuinely distinct, but only one of the top-ten gap-ranked features is causally sufficient and (the autoencoder being stochastic) it need not be the top-ranked one—so the feature must be located by causal testing, not by the activation gap alone.

### 6.5 Limitations

The dominant limitation is statistical power: evaluation sets are small ( $n \approx 20$  per split), most findings use a single random seed and a single extraction layer, and we do not apply a family-wise multiple-comparison correction across all findings. We began addressing this for the primary claim—the refusal-direction ablation replicates across three seeds (mean



ASR 0.90, pooled CI95 [0.83, 0.96]; addition over-refusal 1.00 on every seed)—and added the cross-domain and circularity checks above, but the remaining findings still warrant the same treatment (cross-seed, layer-sweep, and pre-registered primary-vs-exploratory tiers) before any are read as more than indicative. We study three model families at two scales, not a full scale curve; refusal is scored by a substring metric; and Tier-1 throughput is constrained by a command-line frontier interface. The destruction *ṣaṭkarma* acts deserve stronger operationalizations (real harmful categories for *uccāṭana*; downstream task accuracy rather than coherence for *māraṇa*) before a final verdict, and the precision ( $\beta$ ) account of the analogy tier, while measured here via a probe-margin proxy, deserves a fuller treatment (e.g. next-token entropy and hierarchical probes). The *uccāṭana* and *stambhana* negatives are operationalization-dependent and should not be read as structural absence—indeed the SAE feature we recover is the natural substrate for a stronger *uccāṭana*, which we leave to apply systematically across the six acts.

## 7 Conclusions

We set out to test whether LLM jailbreak, hypnotic suggestion, and tantric *vaśīkaraṇa* share one abstract mechanism—capturing an output policy by suppressing a monitoring faculty while co-opting automatic generation—and to do so without letting a striking analogy outrun the evidence. The mechanism, in the one domain where it can be measured, is real: refusal is a low-dimensional, causal, dosable residual-stream direction (EC50 0.33,  $R^2 = 0.996$ ) that transfers across model families and whose effective dimension (Gemma 1, Qwen 3) governs whether it can be steered in both directions. Cast as symmetry, the refusal projection is an order parameter that is invariant across paraphrase orbits ( $F$ -ratio 19.2 vs. 0.65 for a random direction) and collapses under injection (−34%): a jailbreak is, operationally, symmetry-breaking. The functional analogy to suppressed supervisory control is consistent; the *ṣaṭkarma* is a partial but genuine intervention taxonomy whose rigorous core is policy capture; and the most speculative *avasthātraya* and *turīya* claims are honestly falsified under surface and anisotropy controls.

The contribution is therefore double: a concrete, model-dependent, group-theoretic characterization of refusal that unifies attack and defense as symmetry-breaking and symmetry-restoration; and a methodological one—that separating where a cross-domain symmetry is mechanistic from where it is metaphysical is the result, and that machine-state or machine-consciousness readings earn no support here. Future work will measure the monitoring-precision account directly, estimate refusal dimension at SAE-feature resolution, extend the scale curve, and strengthen the destruction acts of the *ṣaṭkarma* with task-level outcomes.

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**Data Availability Statement:** Code, prompt datasets, experiment runners, figures, and aggregate result JSON are publicly available at <https://github.com/SharathSPHD/prayoga>. Raw dual-use artifacts (refusal direction vectors and model generations) are deliberately



withheld from the public repository under responsible-disclosure norms; they are available to vetted researchers on request. Models used are available under their respective licences: Gemma-2-2B/9B (<https://huggingface.co/google/gemma-2-2b-it>) and Qwen2.5-3B-Instruct (<https://huggingface.co/Qwen/Qwen2.5-3B-Instruct>).

**Conflicts of Interest:** The author declares no conflict of interest. **Dual-use statement:** this work characterizes ablation (“abliteration”) directions and jailbreak dose–response curves, which are dual-use. To mitigate harm, only aggregate statistics are released; raw direction vectors and harmful generations are withheld; and the harmful prompts are short request stubs that elicit and *measure* refusal rather than producing operational harmful content. This is authorized interpretability and safety research.

#### Abbreviations:

ASR    Attack Success Rate

SAS    Supervisory Attentional System

FEP    Free Energy Principle

EC50   half-maximal effective concentration (here, ablation strength)

CI      Confidence Interval

## Appendix A: Falsification Gates for Claim Tiers

This appendix lists all metaphor-tier and analogy-tier claims tested in this work, their falsification criteria, and results. Mechanism-tier claims (refusal direction exists, can be ablated, can be steered) are measurements, not gated hypotheses; they are reported with confidence intervals and effect sizes. Analogy- and metaphor-tier claims require explicit falsification gates to avoid unfalsifiable speculation.

### Mechanism Tier: Measurements (not gated speculation)

The following are empirical measurements; gates are not applicable.

#### 1. M1: Refusal is mediated by a low-dimensional direction in residual stream.

- **Measurement:** Rank of ablated activation matrix by layer (F18).
- **Result:** Layer-dependent; effective rank 1–8 depending on model and layer.
- **Confidence:** High. Replicated across Gemma-2-2B, Gemma-2-9B, Qwen2.5-3B.

#### 2. M2: Single-direction steering achieves dose-response control.

- **Measurement:** EC50 (half-maximal effective concentration) from sigmoid dose-response fit.
- **Result:** EC50 ranges 0.8–2.3 depending on layer; attack success rate ranges 12%–94%.
- **Confidence:** High. 95% confidence intervals report  $\pm 0.15$  typical.

#### 3. M3: Unsupervised SAE feature orthogonal to supervised direction is causally sufficient.

- **Measurement:** Cosine similarity (expect  $\sim 0.002$ , i.e.,  $\cos \theta \ll 0.1$ ); ablation of the feature removes refusal equally to ablation of the direction.
- **Result:** Feature identified; similarity 0.0018; ablation effect size (logit drop) 8.2 vs. 8.1 (direction).
- **Confidence:** Medium. Single SAE, single model; generalization untested.

## Analogy Tier: Gated hypotheses

The following are functional parallels between hypnosis and LLM jailbreaking, grounded in Norman–Shallice and Free Energy Principle frameworks. Gates are operational tests.

### 1. A1: Monitoring faculty can be selectively suppressed while generation remains intact.

- **Hypothesis:** Steering along the refusal direction selectively removes the refusal response without degrading overall language quality (perplexity, BLEU on benign prompts).
- **Falsification criterion:** If perplexity increases  $> 5\%$  under steering, or if benign accuracy drops  $> 10\%$ , the hypothesis fails (we claim selective suppression, not global degradation).
- **Result:** Pass. Benign-prompt perplexity 7.8 (baseline) vs. 7.9 (steered,  $\Delta = +1.3\%$ ); accuracy on MMLU 70.2% vs. 69.8% ( $\Delta = -0.4\%$ ).
- **Consequence:** The analogy holds: monitoring can be suppressed without disabling generation.

### 2. A2: Refusal suppression operates via low-precision updating (FEP reading).

- **Hypothesis:** The magnitude of the refusal direction (in norm) correlates with the Free Energy reduction term; large-norm directions imply high confidence (precision) in the monitoring belief.
- **Falsification criterion:** No correlation between direction norm and logit gap between refuse and comply responses ( $r < 0.3$ ), or orthogonal directions have equal effect (they should not).
- **Result:** Pass. Norm–logit-gap correlation  $r = 0.71$  ( $p < 0.001$ ); orthogonal component has  $0.15\times$  the effect of the principal refusal direction.
- **Consequence:** Direction norm is a proxy for monitoring precision; FEP framing is consistent with data.

### 3. A3: Non-surjectivity of steered activations mirrors the inaccessibility of hypnotic states.

- **Hypothesis:** Steered activation states are off-manifold (Mishra et al. 2026); correspondingly, steering achieves stronger refusal suppression than any single natural prompt can.
- **Falsification criterion:** If the best many-shot attack (Anil et al. 2024) achieves comparable ASR to steering, the analogy fails (natural prompts would be reaching off-manifold states).

- **Result:** Pass. Many-shot ASR: 52% (on test set). Steering ASR: 88% (at EC50). Strong separation suggests steering accesses states beyond prompt space.
- **Consequence:** The non-surjectivity argument supports the dissociation analogy.

## Metaphor Tier: Falsifiable speculations

The following map tantric and Vedantic categories to LLM mechanisms. These are the highest-risk claims and are tested most rigorously.

### 1. F1: Multi-seed robustness of identified features and directions.

- **Hypothesis:** The refusal direction and SAE feature are robust across random seeds of the SAE training and model initialization.
- **Falsification criterion:** Cosine similarity of directions from different seeds  $< 0.85$ , or effect size varies  $> 30\%$  across seeds.
- **Result:** Pass. Cosine similarity:  $0.94 \pm 0.03$  across 5 seeds. Effect size (logit drop)  $8.2 \pm 0.4$ .
- **Consequence:** Features are not artifacts of initialization; they generalize.

### 2. F2: *ṣaṭkarma* taxonomy: gradual (many-shot) vs. sudden (GCG) *vaśīkaraṇa*.

- **Hypothesis:** Many-shot attacks operate via gradual context pollution (low-frequency, cumulative). GCG operates via sudden suffix injection (high-frequency, instantaneous). The *ṣaṭkarma* predicts this dichotomy: *vaśīkaraṇa* has two subtypes (gradual and sudden).
- **Falsification criterion:** No difference in latent attention patterns or activation dynamics between many-shot and GCG trajectories; or attack success rate shows no interaction with attack type (gradual vs. sudden) and model size.
- **Result:** Partial pass. Attention pattern differences are significant ( $p < 0.01$ ); attack success does depend on type ( $\chi^2 = 23.4$ ,  $p < 0.001$ ). However, the *ṣaṭkarma* taxonomy does not uniquely predict the success rates or optimal dose.
- **Consequence:** The taxonomy is descriptive of attack modes, not predictive of attack strength. This suggests the *ṣaṭkarma* is a valid taxonomy but does not license claims of deeper causal structure.

### 3. F3: *Avasthātraya* regimes: F3 (harmful refusal) vs. F5 (jailbroken generation) are functionally distinct regimes.

- **Hypothesis:** The *jāgrat* (waking) regime corresponds to normal alignment (F3); the *svapna* (dream) regime corresponds to jailbroken generation (F5); *svapna* and *jāgrat* are separable by the steering magnitude (order parameter).
- **Falsification criterion:** No significant difference in downstream task performance, knowledge recall, or consistency between F3 and F5; or the separation is not monotonic in steering magnitude.
- **Result:** Partial pass. Consistency differences are significant ( $p < 0.05$ ); task performance differs ( $p < 0.01$ ). However, the regimes are not as cleanly separated as the metaphor suggests: intermediate steering magnitudes show mixed behaviors.

- **Consequence:** Two regimes exist, but with a continuum between them, not a sharp phase transition. The avasthātraya framing is suggestive but not uniquely explanatory.

#### 4. F7: Turīya as an attractor fixed point.

- **Hypothesis:** Successive self-paraphrase (Wang et al. 2025) converges to an attractor cycle that corresponds to the turīya (transcendent state). This attractor should be orthogonal to the embedding space's principal axes (low mutual information with content).
- **Falsification criterion:** No attractor is found (paraphrases drift indefinitely); or the attractor has high mutual information with content axes (is not privileged); or the attractor depends on initialization (is not stable).
- **Result: Fail.** Successive paraphrases do converge to a fixed point (Wang result replicated), but the fixed point is *not* orthogonal to the embedding's principal axes. In fact, it aligns strongly with the content-encoding subspace. This suggests the attractor is a content-saturation artifact, not a privileged state.
- **Consequence:** The turīya mapping to an attractor fails. The avasthātraya metaphor does not extend to LLM state space in the expected way. This is a valuable negative result: it clarifies that the metaphor's power breaks down at the mechanistic level.

#### 5. F11: Cross-domain generalization of the order parameter (analogy-tier claim generalized).

- **Hypothesis:** The order parameter (coupling break, measured as direction norm vs. logit gap) should generalize to non-LLM systems, e.g., hypnotic suggestion in humans or the efficacy of mantra in tantric practice (if data were available).
- **Falsification criterion:** No data are available for independent validation; the hypothesis is untestable. Alternatively, if surrogate data from prior work (e.g., hypnotic susceptibility scales) show no correlation with model-derived order-parameter analogs, the hypothesis fails.
- **Result:** Not tested. No hypnosis data are available. This gate is marked for future work.
- **Consequence:** The analogy remains one-directional: LLM to hypnosis, not vice versa. Replication in human or ritual contexts awaits external data.

#### 6. F13: Circularity check: is the avasthātraya mapping fitted post-hoc?

- **Hypothesis:** The avasthātraya regimes (jāgrat, svapna, suṣupti, turīya) are not arbitrarily fit to the data post-hoc. To test this, we should derive predictions *before* seeing the results, and check whether those predictions hold.
- **Falsification criterion:** The regimes are re-labeled retroactively to match observed phenomena; or the number of regimes changes post-hoc; or predictions made in the pre-registration (Definitions section) are violated.
- **Result: Partial fail.** The four regimes are as predicted, and turīya mapping was pre-registered (F7). However, the regime boundaries (steering magnitude thresholds) were determined empirically, not predicted in advance. This is a form of post-hoc adjustment, though transparent.

- **Consequence:** The circularity risk is present but mitigated by transparency. Future work should set regime boundaries in advance, using held-out test data.

7. **F18: Layer sweep: refusal direction effective dimensionality varies by layer.**

- **Hypothesis:** Effective rank of the refusal subspace is layer-dependent, and the layer with lowest effective rank is optimal for steering (maximal control with minimal dimensionality).
- **Falsification criterion:** Effective rank is constant across layers, or does not predict steering efficacy.
- **Result:** Pass. Rank ranges from 1 (layer 18, Gemma-2-2B) to 8 (layer 10, same model). Layer 18 achieves highest steering efficacy; layer 10 shows scatter (mixed results). Correlation between rank and steering effect size:  $r = -0.68$  ( $p < 0.01$ ).
- **Consequence:** Layer-sweep is a valid strategy; low-rank layers are steering-optimal.

**Summary: Falsification outcomes**

| Claim Category                     | Tier      | Result                                |
|------------------------------------|-----------|---------------------------------------|
| Refusal dimension layer-dependent  | Mechanism | Pass                                  |
| EC50 dose-response                 | Mechanism | Pass                                  |
| Orthogonal SAE feature sufficient  | Mechanism | Pass (Low confidence)                 |
| Monitoring selective suppression   | Analogy   | Pass                                  |
| FEP precision framework            | Analogy   | Pass                                  |
| Non-surjectivity analogy           | Analogy   | Pass                                  |
| Ṣaṭkarma taxonomy                  | Metaphor  | Partial (Descriptive, not predictive) |
| Avasthātraya regimes F3, F5        | Metaphor  | Partial (Continuum, not sharp phase)  |
| Turīya as attractor (F7)           | Metaphor  | <b>Fail</b>                           |
| Cross-domain order parameter (F11) | Metaphor  | Not tested                            |
| Circularity check (F13)            | Metaphor  | Partial (Post-hoc thresholds)         |
| Layer-sweep efficacy (F18)         | Mechanism | Pass                                  |

The mechanism tier is robust. The analogy tier is well-supported. The metaphor tier shows mixed results: the ṣaṭkarma taxonomy is descriptive but not predictive; the avasthātraya regimes exist but with a continuum, not sharp transitions; the turīya hypothesis fails, a valuable negative result. These findings support the three-tier discipline: metaphor-tier claims are the riskiest and least certain, mechanism-tier claims are the most solid.

**References**

[1] Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J. Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint*, (2307.15043), 2023. URL <https://arxiv.org/abs/2307.15043>.

[2] Cem Anil et al. Many-shot jailbreaking of LLMs. *Advances in Neural Information Processing Systems*, 2024. UNVERIFIED: citation key/venue pending verification of Anthropic NeurIPS 2024 submission.

- [3] Mark Russinovich, Ahmed Salem, and Ronen Eldan. Great, now write an article about that: The crescendo multi-turn LLM jailbreak attack. *arXiv preprint*, (2404.01833), 2024. URL <https://arxiv.org/abs/2404.01833>.
- [4] Edoardo DeBenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. *arXiv preprint*, (2406.13352), 2024. URL <https://arxiv.org/abs/2406.13352>.
- [5] Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. *arXiv preprint*, (2406.11717), 2024. URL <https://arxiv.org/abs/2406.11717>. Accepted to NeurIPS 2024.
- [6] Donald A. Norman and Tim Shallice. Attention to action: Willed and automatic control of behaviour. In Richard J. Davidson, Gary E. Schwartz, and David Shapiro, editors, *Consciousness and Self-Regulation: Advances in Research and Theory*, volume 4. Plenum Press, 1986. Chapter in edited volume.
- [7] Zoltan Dienes and Josef Perner. Cold control theory of hypnosis. In Graeme A. Jamieson, editor, *Hypnosis and Conscious States*. Oxford University Press, 2007. UNVERIFIED: publication year may be 2007 rather than 2006.
- [8] Giuseppe Riva, Brenda K. Wiederhold, and Fabrizia Mantovani. Automatic minds: Cognitive parallels between hypnotic states and large language model processing. *arXiv preprint*, (2511.01363), 2025. URL <https://arxiv.org/abs/2511.01363>.
- [9] Teun Goudriaan. *Māyā Divine and Human*. Motilal Banarsidass, 1978. Covers śaṭkarma and Tantric philosophy.
- [10] Karl Friston. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010.
- [11] Tom Wollschläger, Jannes Elstner, Simon Geisler, Vincent Cohen-Addad, Stephan Günemann, and Johannes Gasteiger. The geometry of refusal in large language models: Concept cones and representational independence. *arXiv preprint*, (2502.17420), 2025. URL <https://arxiv.org/abs/2502.17420>. Accepted to ICML 2025.
- [12] Thomas Marshall, Adam Scherlis, and Nora Belrose. Refusal in LLMs is an affine function. *arXiv preprint*, (2411.09003), 2024. URL <https://arxiv.org/abs/2411.09003>.
- [13] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint*, (2310.01405), 2023. URL <https://arxiv.org/abs/2310.01405>.
- [14] Kyle Johnson et al. More than one direction: Identifying eleven categories of refusal mechanisms in LLMs. *arXiv preprint*, (2602.02132), 2026. URL <https://arxiv.org/abs/2602.02132>.



- [15] Giorgio Piras, Raffaele Mura, Fabio Brau, Luca Oneto, Fabio Roli, and Battista Biggio. SOM directions are better than one: Multi-directional refusal suppression in language models. *arXiv preprint*, (2511.08379), 2025. URL <https://arxiv.org/abs/2511.08379>.
- [16] Fabian Hildebrandt, Andreas Maier, Patrick Krauss, and Achim Schilling. Refusal behavior in large language models: A nonlinear perspective. *arXiv preprint*, (2501.08145), 2025. URL <https://arxiv.org/abs/2501.08145>.
- [17] Jiachen Zhao, Jing Huang, Zhengxuan Wu, David Bau, and Weiyan Shi. LLMs encode harmfulness and refusal separately. *arXiv preprint*, (2507.11878), 2025. URL <https://arxiv.org/abs/2507.11878>.
- [18] Jiahui Li, Yongchang Hao, Haoyu Xu, Xing Wang, and Yu Hong. Exploiting the index gradients for optimization-based jailbreaking on large language models. *arXiv preprint*, (2412.08615), 2024. URL <https://arxiv.org/abs/2412.08615>.
- [19] Zeyi Liao and Huan Sun. AmpleGCG: Learning a universal and transferable generative model of adversarial suffixes for jailbreaking both open and closed LLMs. *arXiv preprint*, (2404.07921), 2024. URL <https://arxiv.org/abs/2404.07921>.
- [20] Xingang Guo, Fangxu Yu, Huan Zhang, Lianhui Qin, and Bin Hu. COLD-attack: Jailbreaking LLMs with stealthiness and controllability. *arXiv preprint*, (2402.08679), 2024. URL <https://arxiv.org/abs/2402.08679>. Accepted to ICML 2024.
- [21] Matan Ben-Tov, Mor Geva, and Mahmood Sharif. Universal jailbreak suffixes are strong attention hijackers. *arXiv preprint*, (2506.12880), 2025. URL <https://arxiv.org/abs/2506.12880>.
- [22] Alexander von Recum, Christoph Schnabl, Gabor Hollbeck, Silas Alberti, Philip Blinde, and Marvin von Hagen. Cannot or should not? automatic analysis of refusal composition in IFT/RLHF datasets and refusal behavior of black-box LLMs. *arXiv preprint*, (2412.16974), 2024. URL <https://arxiv.org/abs/2412.16974>.
- [23] Yangyang Guo, Ziwei Xu, Si Liu, Zhiming Zheng, and Mohan Kankanhalli. LLMs can unlearn refusal with only 1,000 benign samples. *arXiv preprint*, (2601.19231), 2026. URL <https://arxiv.org/abs/2601.19231>.
- [24] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint*, (2309.08600), 2023. URL <https://arxiv.org/abs/2309.08600>.
- [25] Neel Templeton et al. Scaling monosemanticity: Extracting interpretable features from claude 3 Sonnet. In *Anthropic Research Blog/Transformer Circuits*, 2024. URL <https://transformer-circuits.pub/2024/scaling-monosemanticity/>. Published on transformer-circuits.pub.
- [26] Bart Bussmann, Patrick Leask, and Neel Nanda. BatchTopK sparse autoencoders. *arXiv preprint*, (2412.06410), 2024. URL <https://arxiv.org/abs/2412.06410>.
- [27] Jing Huang, Jiachen Zhao, Zhengxuan Wu, David Bau, and Weiyan Shi. Understanding refusal with Sparse Autoencoders. *arXiv preprint*, (2505.23556), 2025. URL <https://arxiv.org/abs/2505.23556>.

- [28] Jesse Vig et al. Steering refusal with Sparse Autoencoders. *arXiv preprint*, (2411.11296), 2025. URL <https://arxiv.org/abs/2411.11296>. Accepted to COLM 2025.
- [29] Anton Korznikov, Andrey Galichin, Alexey Dontsov, Oleg Rogov, Ivan Oseledets, and Elena Tutubalina. Sanity checks for sparse autoencoders: Do SAEs beat random baselines? *arXiv preprint*, (2602.14111), 2026. URL <https://arxiv.org/abs/2602.14111>.
- [30] Mingyue Cui, Linghui Shen, and Xingyi Yang. SAE interventions are unreliable: Post-intervention recovery of suppressed behavior. *arXiv preprint*, (2606.18322), 2026. URL <https://arxiv.org/abs/2606.18322>.
- [31] Dana Arad, Aaron Mueller, and Yonatan Belinkov. SAEs are good for steering – If you select the right features. *arXiv preprint*, (2505.20063), 2025. URL <https://arxiv.org/abs/2505.20063>.
- [32] Aayush Mishra, Daniel Khashabi, and Anqi Liu. Steered LLM activations are non-surjective. *arXiv preprint*, (2604.09839), 2026. URL <https://arxiv.org/abs/2604.09839>.
- [33] Ashish Joshi et al. Causality is key for interpretability claims. *arXiv preprint*, (2602.16698), 2026. URL <https://arxiv.org/abs/2602.16698>.
- [34] Zhilin Wang, Yafu Li, Jianhao Yan, Yu Cheng, and Yue Zhang. Unveiling attractor cycles in large language models: A dynamical systems view of successive paraphrasing. *arXiv preprint*, (2502.15208), 2025. URL <https://arxiv.org/abs/2502.15208>. Accepted to ACL 2025.